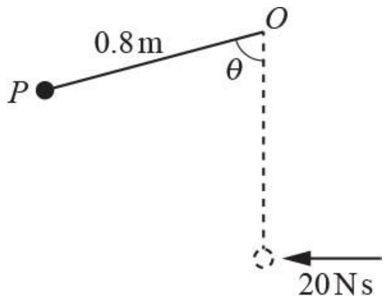
OCR Further Mathematics
Mechanics - Work, Energy and Power
Question Paper
AS and A Level
Further Mathematics A - H235, H245

The latest topic questions for the year 2024 / 2025+ exams.
To be used by all students preparing for OCR AS and A Level Further Mathematics.
Students of other boards may also find this useful.

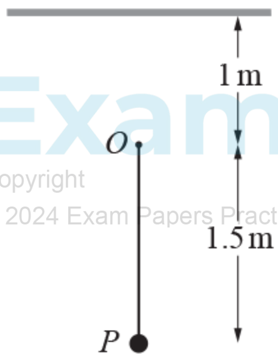
1(a) A particle P of mass 4 kg is attached to one end of a light inextensible string of length 0.8 m. The other end of the string is attached to a fixed point O. P is at rest vertically below O when it experiences a horizontal impulse of magnitude 20 N s. In the subsequent motion the angle the string makes with the downwards vertical through O is denoted by θ (see diagram).



Find the magnitude of the acceleration of P at the first instant when $\theta = \frac{1}{3}\pi$ radians. [7]

(b) Determine the value of θ at which the string first becomes slack. [3]

2 One end of a light elastic string of natural length 0.6 m and modulus of elasticity 24 N is attached to a fixed point O. The other end is attached to a particle P of mass 0.4 kg. O is a vertical distance of 1 m below a horizontal ceiling. P is held at a point 1.5 m vertically below O and released from rest (see diagram).



Assuming that there is no obstruction to the motion of P as it passes O, find the speed of P when it first hits the ceiling. [5]

3(a) One end of a light elastic string of natural length l m and modulus of elasticity λ N is attached to a particle A of mass m kg. The other end of the string is attached to a fixed point O which is on a horizontal surface. The surface is modelled as being smooth and A moves in a circular path around O with constant speed v ms^{-1}. The extension of the string is denoted by x m.

Show that x satisfies $\lambda x^2 + \lambda l x - lmv^2 = 0$. [3]

(b) By solving the equation in part (a) and using a binomial series, show that if λ is very large then $\lambda x \approx mv^2$. [3]

(c) By considering the tension in the string, explain how the result obtained when λ is very large relates to the situation when the string is inextensible. [1]

(d) The nature of the horizontal surface is such that the modelling assumption that it is smooth is justifiable provided that the speed of the particle does not exceed 7ms^{-1}.

In the case where $m = 0.16$ and $\lambda = 260$, the extension of the string is measured as being 3.0 cm.

Estimate the value of v. [1]

(e) Explain whether the value of v means that the modelling assumption is necessarily justifiable in this situation. [1]

4(a) One end of a light inextensible string of length 0.75 m is attached to a particle A of mass 2.8 kg. The other end of the string is attached to a fixed point O. A is projected horizontally with speed 6ms^{-1} from a point 0.75 m vertically above O (see Fig. 3). When OA makes an angle θ with the upward vertical the speed of A is v ms^{-1}.

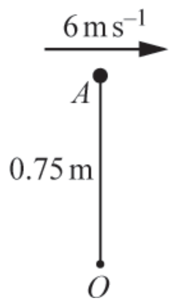


Fig. 3

Show that $v^2 = 50.7 - 14.7 \cos \theta$. [3]

(b) Given that the string breaks when the tension in it reaches 200 N, find the angle that *OA* turns through between the instant that *A* is projected and the instant that the string breaks. [4]

5 A bungee jumper of mass 80 kg steps off a high bridge with an elastic rope attached to her ankles. She is assumed to fall vertically from rest and the air resistance she experiences is modelled as a constant force of 32 N. The rope has natural length 4 m and modulus of elasticity 470 N.

By considering energy, determine the total distance she falls before first coming to instantaneous rest. [6]

6(a)

A force of $\begin{pmatrix} 2 \\ 10 \end{pmatrix}$ N is the only horizontal force acting on a particle P of mass 1.25 kg as it moves in a horizontal plane. Initially P is at the origin, O, and 5 seconds later it is at the point A (50, 140). The units of the coordinate system are metres.

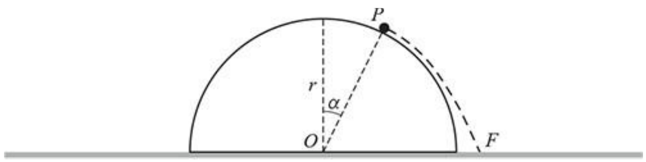
Calculate the work done by the force during these 5 seconds. [2]

(b) Calculate the average power generated by the force during these 5 seconds. [1]

(c) The speed of P at O is 10 ms^{-1}.

Calculate the speed of P at A. [2]

7(a)



The flat surface of a smooth solid hemisphere of radius r is fixed to a horizontal plane on a planet where the acceleration due to gravity is denoted by γ. O is the centre of the flat surface of the hemisphere.

A particle P is held at a point on the surface of the hemisphere such that the angle between OP and the upward vertical through O is α, where $\cos\alpha = \frac{3}{4}$.

P is then released from rest. F is the point on the plane where P first hits the plane (see diagram).

Find an exact expression for the distance OF. [11]

(b) The acceleration due to gravity on and near the surface of the planet Earth is roughly 6γ.

Explain whether OF would increase, decrease or remain unchanged if the action were repeated on the planet Earth. [1]

8(a) Two particles A and B, of masses m kg and 1 kg respectively, are connected by a light inextensible string of length d m and placed at rest on a smooth horizontal plane a distance of $\frac{1}{2}$ dm apart. B is then projected horizontally with speed $v$ms^{-1} in a direction perpendicular to AB.

Show that, at the instant that the string becomes taut, the magnitude of the instantaneous impulse in the string, I N s, is given by $I = \frac{\sqrt{3}mv}{2(1+m)}$.

[4]

(b) Find, in terms of m and v, the kinetic energy of B at the instant after the string becomes taut. Give your answer as a single algebraic fraction. [3]

(c) In the case where m is very large, describe, with justification, the approximate motion of B after the string becomes taut. [2]

9(a) A right circular cone C of height 4 m and base radius 3 m has its base fixed to a horizontal plane. One end of a light elastic string of natural length 2 m and modulus of elasticity 32 N is fixed to the vertex of C. The other end of the string is attached to a particle P of mass 2.5 kg.

P moves in a horizontal circle with constant speed and in contact with the smooth curved surface of C. The extension of the string is 1.5 m.

Find the tension in the string. [2]

(b) Find the speed of P. [7]

10(a) A particle Q of mass m kg is acted on by a single force so that it moves with constant acceleration

$\mathbf{a} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \mathrm{m\,s}^{-2}$. Initially Q is at the point O and is moving with velocity $\mathbf{u} = \begin{pmatrix} 2 \\ -5 \end{pmatrix} \mathrm{m\,s}^{-1}$.

After Q has been moving for 5 seconds it reaches the point A.

Use the equation v.v = u.u+2a.x to show that at A the kinetic energy of Q is 37m J. [5]

(b) (i) Show that the power initially generated by the force is –8*m* W. [2]

(ii) The power in part (i) is negative. Explain what this means about the initial motion of *Q*. [1]

(c) (i) Find the time at which the power generated by the force is instantaneously zero. [3]

(ii) Find the minimum kinetic energy of Q in terms of m. [2]

11 Particles A, B and C of masses 2 kg, 3 kg and 5 kg respectively are joined by light rigid rods to form a triangular frame. The frame is placed at rest on a horizontal plane with A at the point (0, 0), B at the point (0.6, 0) and C at the point (0.4, 0.2), where distances in the coordinate system are measured in metres (see Fig. 1).

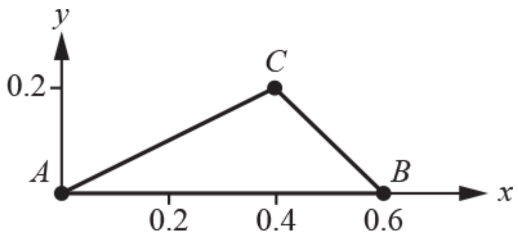


Fig. 1

G, which is the centre of mass of the frame, is at the point $(\bar{x}, \bar{y})$.

(a) • Show that $\bar{x} = 0.38$,

• Find $\bar{y}$. [3]

(b) Explain why it would be impossible for the frame to be in equilibrium in a horizontal plane supported at only one point. [1]

A rough plane, *Π*, is inclined at an angle i to the horizontal where $\sin\theta = \frac{3}{5}$. The frame is placed on *Π* with *AB* vertical and *B* in contact with *Π*. *C* is in the same vertical plane as *AB* and a line of greatest slope of *Π*. *C* is on the down-slope side of *AB*.

The frame is kept in equilibrium by a horizontal light elastic string whose natural length is *l* m and whose modulus of elasticity is g N. The string is attached to A at one end and to a fixed point on *Π* at the other end (see Fig. 2).

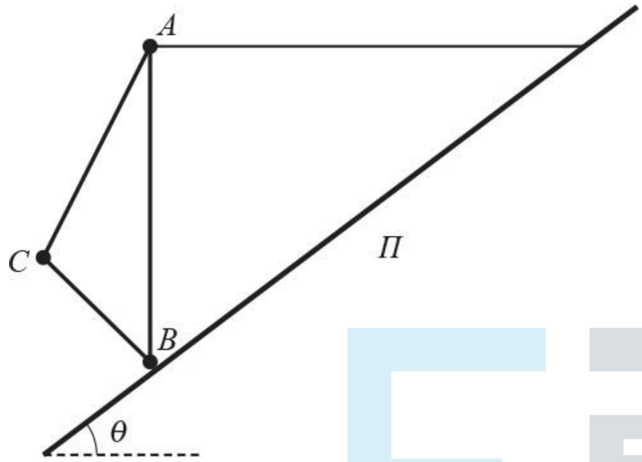


Fig. 2

The coefficient of friction between *B* and *Π* is *μ*.

(c) Show that *l* = 0.3. [3]

(d) Show that $\mu \geqslant \frac{14}{27}$. [6]

12 A car of mass 800 kg is driven with its engine generating a power of 15 kW.

(a) The car is first driven along a straight horizontal road and accelerates from rest.
Assuming that there is no resistance to motion, find the speed of the car after 6 seconds. [2]

(b) The car is next driven at constant speed up a straight road inclined at an angle θ to the horizontal.
The resistance to motion is now modelled as being constant with magnitude 150 N.

Given that $\sin\theta = \frac{1}{20}$, find the speed of the car. [3]

(c) The car is now driven at a constant speed of $30ms^{-1}$ along the horizontal road pulling a trailer of mass 150 kg which is attached by means of a light rigid horizontal towbar.

Assuming that the resistance to motion of the car is three times the resistance to motion of the trailer, find

- the resistance to motion of the car,
- the magnitude of the tension in the towbar. [4]

13

A particle, P, of mass 2 kg moves in two dimensions. Its initial velocity is $\begin{pmatrix} -19.5 \\ -60 \end{pmatrix} \mathrm{m\,s^{-1}}$.

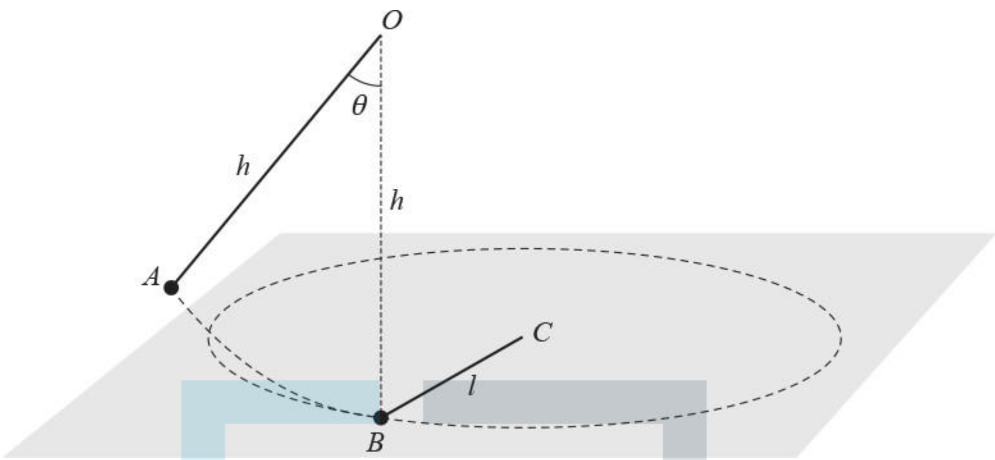
(a) Calculate the initial kinetic energy of p. [2]

For $t \geq 0$, P is acted upon only by a variable force $\mathbf{F} = \begin{pmatrix} 4t \\ -2 \end{pmatrix} \mathrm{N}$, where t is the time in seconds.

(b) Find

- the velocity of P in terms of t,
- the times when the power generated by F is zero. [6]

14 A point O is situated a distance h above a smooth horizontal plane, and a particle A of mass m is attached to O by a light inextensible string of length h. A particle B of mass $2m$ is at rest on the plane, directly below O, and is attached to a point C on the plane, where $BC = l$, by a light inextensible string of length l. A is released from rest with the string OA taut and making an acute angle θ with the downward vertical (see diagram).



A moves in a vertical plane perpendicular to CB and collides directly with B. As a result of this collision, A is brought to rest and B moves on the plane in a horizontal circle with centre C. After B has made one complete revolution the particles collide again.

(a) Show that, on the next occasion that A comes to rest, the string OA makes an angle ϕ with the downward vertical through O, where $\cos\phi = \dfrac{3+\cos\theta}{4}$. [9]

A and *B* collide again when *AO* is next vertical.

(b) Find the percentage of the original energy of the system that remains immediately after this collision. [5]

(c) Explain why the total momentum of the particles immediately before the first collision is the same as the total momentum of the particles immediately after the second collision. [1]

(d) Explain why the total momentum of the particles immediately before the first collision is different from the total momentum of the particles immediately after the third collision. [1]

15 A particle P of mass m moves along the positive x-axis. When its displacement from the origin O is x its velocity is v, where $v \geq 0$. It is subject to two forces: a constant force T in the positive x direction, and a resistive force which is proportional to v^2.

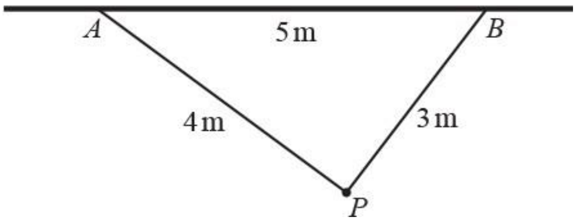
(a) Show that $v^2 = \frac{1}{k}\left(T - Ae^{-\frac{2kx}{m}}\right)$ where A and k are constants. [5]

P starts from rest at O.

(b) Find an expression for the work done against the resistance to motion as P moves from O to the point where $x = 1$. [4]

(c) Find an expression for the limiting value of the velocity of P as x increases. [1]

16 *A* and *B* are two points a distance of 5 m apart on a horizontal ceiling. A particle *P* of mass *m* kg is attached to *A* and *B* by light elastic strings. The particle hangs in equilibrium at a distance of 4 m from *A* and 3 m from *B* so that angle *APB* = 90° (see diagram).



The string joining *P* to *A* has natural length 2 m and modulus of elasticity λ_A N. The string joining P to B also has natural length 2 m but has modulus of elasticity λ_B N.

(a) (i) Show that $\lambda_B = \frac{8}{3}\lambda_A$. [4]

(ii) Find an expression for λ_A in terms of *m* and *g*. [3]

(b) Find, in terms of m and g, the total elastic potential energy stored in the strings. [2]

The string joining P to A is detached from A and a second particle, Q, of mass $0.3\,m$ kg is attached to the free end of the string. Q is then gently lowered into a position where the system hangs vertically in equilibrium.

(c) Find the distance of Q below B in this equilibrium position. [4]

17 A car of mass 850 kg is being driven uphill along a straight road inclined at 7° to the horizontal. The esistance to motion is modelled as a constant force of magnitude 140 N. At a certain instant the car's speed is 12ms^{-1} and its acceleration is 0.4ms^{-2}.

(a) Calculate the power of the car's engine at this instant. [3]

(b) Find the constant speed at which the car could travel up the hill with the engine generating this power. [2]

18 A particle P of mass 2.5 kg strikes a rough horizontal plane. Immediately before P strikes the plane it has a speed of 6.5 m s^{-1} and its direction of motion makes an angle of 30° with the normal to the plane at the point of impact. The impact may be assumed to occur instantaneously. The coefficient of restitution between P and the plane is $\frac{2}{3}$. The friction causes a horizontal impulse of magnitude 2 N s to be applied to P in the plane in which it is moving.

(a) Calculate the velocity of P immediately after the impact with the plane. [7]

(b) P loses about x % of its kinetic energy as a result of the impact. Find the value of x. [2]

19 A ball B of mass 1.7 kg is connected to one end of a light elastic spring of natural length 1.2 m. The other end of the spring is attached to a point O on the ceiling of a large room. The modulus of elasticity of the spring is 50 N. The ball is held 3.2 m vertically below O and projected upwards with an initial speed of 0.5 m s^{-1}. In order to model the motion of B (before any collision with the ceiling) the following assumptions are made.

- Air resistance is ignored.
- B is small.
- The fully compressed length of the spring is negligible.

(a) Determine whether, according to the model, B reaches O. [4]

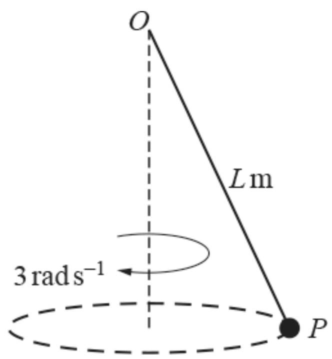
(b) Without doing any further calculations, explain whether the answer to part (a) could change in each of the following different cases.

(i) A new model is used in which air resistance is taken into account. [1]

(ii) The spring is replaced by an elastic string with the same natural length and modulus of elasticity. [1]

(iii) *B* is initially projected downwards rather than upwards. [1]

20 A particle P of mass 3.5 kg is attached to one end of a light elastic string of natural length 0.8 m and modulus of elasticity 75 N. The other end of the string is attached to a fixed point O. The particle rotates in a horizontal circle with a constant angular velocity of 3 rad s^{-1}. The centre of the circle is vertically below O. The magnitude of the tension in the string is T N and the length of the extended string is L m (see diagram).



(a) By considering the acceleration of P, show that $T = 31.5L$. [4]

(b) Write down another relationship between T and L. [1]

(c) Find the value of T and the value of L. [3]

(d) Find the angle that the string makes with the downwards vertical through O. [2]

21 A particle P of mass 4.2 kg is free to move along the x-axis which is horizontal. P is projected from the origin, O, in the positive x direction with a speed of 2 m s^{-1}. As P moves between O and the point A where $x = 4$, it is acted upon by a variable force of magnitude $(12x - 3x^2)$ N acting in the direction OA.

(a) Calculate the work done by the force as P moves from O to A. [2]

(b) Hence, assuming that no other force acts on P, calculate the speed of P at A. [4]

22 A car of mass 1200 kg travels up a line of greatest slope of a straight road inclined at 4° to the horizontal. The power of the car's engine is constant and equal to 23 kW and the resistance to the motion of the car is constant and equal to 800 N. The car passes through a point *A* on the road with speed 8 m s^{-1}.

(i) Find

(a) the acceleration of the car at *A*,

(b) the greatest steady speed at which the car can travel up the hill. [5]

The car later passes through a point B on the same road where $AB = 109$ m and the car takes 10.1 s to travel from A to B.

(ii) Calculate the speed of the car at B. [7]

23 A boy drags a sledge in a straight line along horizontal ground by means of a rope attached to the sledge. The rope makes an angle of 25° with the horizontal and the tension in the rope, T N, is constant. The work done by the tension in moving the sledge 75 m is 4000 J. Calculate the value of T. [3]

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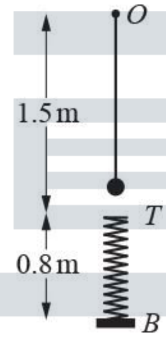

A particle of mass m kg is attached to one end of a light elastic string of natural length 1.2 m and modulus of elasticity $24mg$ N. The other end of the string is attached to a fixed point O. Directly beneath O there is a light elastic spring, of natural length 0.8 m and modulus of elasticity $32mg$ N. The bottom of the spring is attached to a fixed point B and the top of the spring is at a point T, 1.5 m vertically below O; the spring is constrained to remain vertical. The particle is projected from O with speed 0.7 m s^{-1} vertically downwards. When the particle reaches T it becomes attached to the spring and it remains attached to the spring throughout the subsequent motion. The diagram shows the position as the particle first approaches T.

(i) Show that the speed of the particle at the instant when it becomes attached to the spring is 3.5 m s^{-1}. [3]

(ii) Find the distances below O at which the particle is instantaneously at rest in the subsequent motion. [5]

25 A light inextensible taut rope, of length 4 m, is attached at one end A to the centre of the horizontal ceiling of a gym. The other end of the rope B is being held by a child of mass 35 kg. Initially the child is held at rest with the rope making an angle of 60° to the downward vertical and it may be assumed that the child can be modelled as a particle attached to the end of the rope. The child is released at a height 5 m above the horizontal ground.

(a) Show that the speed, v m s^{-1}, of the child when the rope makes an angle θ with the downward vertical

is given by $v^2 = 4g(2\cos\theta - 1)$.

[5]

(b) At the instant when $\theta = 0°$, the child lets go of the rope and moves freely under the influence of gravity only. Determine the speed and direction of the child at the moment that the child reaches the ground. [5]

(c) The child returns to the initial position and is released again from rest. Find the value of θ when the tension in the rope is three times greater than the tension in the rope at the instant the child is released.

[5]

26 As part of a training exercise an army recruit of mass 75 kg falls a vertical distance of 5 m before landing on a mat of thickness 1.2 m. The army recruit sinks a distance of x m into the mat before instantaneously coming to rest. The mat can be modelled as a spring of natural length 1.2 m and modulus of elasticity 10 800 N and the army recruit can be modelled as a particle falling vertically with an initial speed of 2 m s^{-1}.

(a) Show that x satisfies the equation $300x^2 - 49x - 255 = 0$. [5]

(b) Calculate the value of x. [2]

(c) Ignoring the possible effect of air resistance, make

- one comment on the assumptions made and,
- suggest a possible refinement to the model. [2]

27 A body, P, of mass 2 kg moves under the action of a single force F N. At time t s, the velocity of the body is v m s^{-1}, where

$$\mathbf{v} = \left(t^2 - 3\right)\mathbf{i} + \frac{5}{2t+1}\mathbf{j} \text{ for } t \geq 2.$$

(a) Obtain F in terms of t. [3]

(b) Calculate the rate at which the force F is working at $t = 4$. [3]

(c) By considering the change in kinetic energy of P, calculate the work done by the force F during the time interval $2 \leq t \leq 4$. [3]

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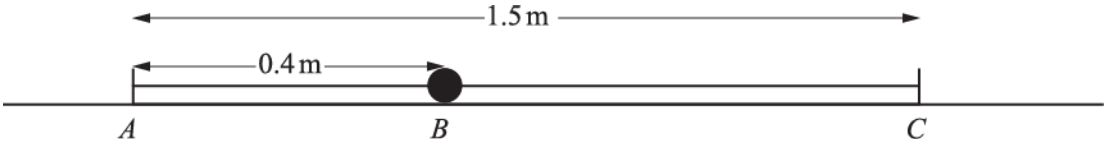


A and *C* are two fixed points, 1.5 m apart, on a smooth horizontal plane. A light elastic string of natural length 0.4 m and modulus of elasticity 20 N has one end fixed to point *A* and the other end fixed to a particle *B*. Another light elastic string of natural length 0.6 m and modulus of elasticity 15 N has one end fixed to point *C* and the other end fixed to the particle *B*. The particle is released from rest when *ABC* forms a straight line and *AB* = 0.4 m (see diagram).

Find the greatest kinetic energy of particle *B* in the subsequent motion.

29 A particle *P* of mass *m* kg is attached to one end of a light elastic string of modulus of elasticity 24 *mg* N and

[7]

natural length 0.6 m. The other end of the string is attached to a fixed point *O*; the particle *P* rests in equilibrium at a point *A* with the string vertical.

(i) Show that the distance *OA* is 0.625 m.

[2]

Another particle Q, of mass $3m$ kg, is released from rest from a point 0.4 m above P and falls onto P. The two particles coalesce.

(ii) Show that the combined particle initially moves with speed 2.1 m s^{-1}.

[3]

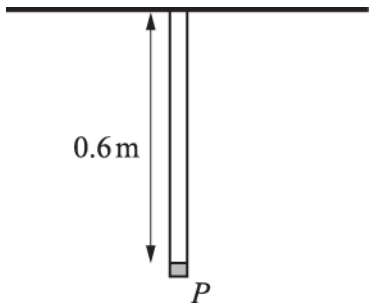
(iii) Show that the combined particle initially performs simple harmonic motion, and find the centre of this motion and its amplitude.

[5]

(iv) Find the time that elapses between Q being released from rest and the combined particle first reaching the highest point of its subsequent motion.

[7]

30



A small object P is attached to one end of each of two vertical light elastic strings. One string is of natural length 0.4 m and has modulus of elasticity 10 N; the other string is of natural length 0.5 m and has modulus of elasticity 12 N. The upper ends of both strings are attached to a fixed horizontal beam and P hangs in equilibrium 0.6 m below the beam (see diagram).

(i) Show that the weight of P is 7.4 N and find the total elastic potential energy stored in the two strings when P is hanging in equilibrium.

[6]

P is then held at a point 0.7 m below the beam with the strings vertical. P is released from rest.

(ii) Show that, throughout the subsequent motion, P performs simple harmonic motion, and find the period.

[7]

31 A smooth cylinder of radius a m is fixed with its axis horizontal and O is the centre of a cross-section. Particle P, of mass 0.4 kg, and particle Q, of mass 0.6 kg, are connected by a light inextensible string of length πa m. The string is held at rest with P and Q at opposite ends of the horizontal diameter of the cross-section through O (see Fig. 1). The string is released and Q begins to descend. When OP has rotated through i radians, with P remaining in contact with the cylinder, the speed of each particle is v m s^{-1} (see Fig. 2).

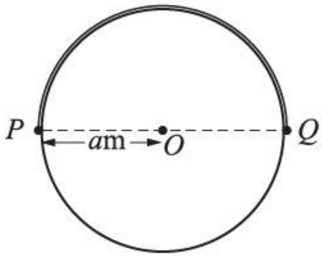


Fig. 1

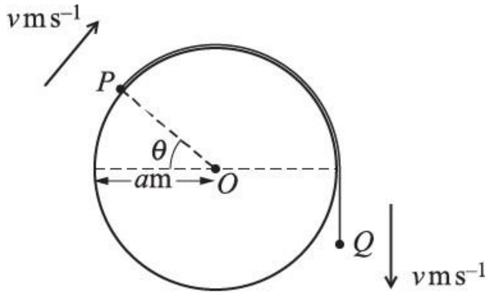


Fig. 2

(i) Show that $v^2 = 3.92a\,(3\theta - 2\sin\theta)$ and find an expression in terms of θ for the normal force of the cylinder on P at this time.

[9]

(ii) Given *that* P leaves the surface of the cylinder when $\theta = \alpha$, show that $\sin\alpha = k\alpha$ where k is a constant to be found.

[2]

32 A particle P, of mass 2.5 kg, is in equilibrium suspended from a fixed point A by a light elastic string of natural length 3 m and modulus of elasticity 36.75 N. Another particle Q, of mass 1 kg, is released from rest at A and falls freely until it reaches P and becomes attached to it.

(i) Show that the speed of the combined particles, immediately after Q becomes attached to P, is $2\sqrt{2}\ \mathrm{m\,s^{-1}}$.

[6]

The combined particles fall a further distance X m before coming to instantaneous rest.

(i) Find a quadratic equation satisfied by X, and show that it simplifies to $35X^2 - 56X - 80 = 0$.

[6]

33 One end of a light elastic string, of natural length 0.6 m and modulus of elasticity 30 N, is attached to a fixed point O. A particle P of weight 48 N is attached to the other end of the string. P is released from rest at a point d m vertically below O. Subsequently P just reaches O.

(i) Find d.

[4]

(ii) Find the magnitude and direction of the acceleration of *P* when it has travelled 1.3 m from its point of release.

[4]

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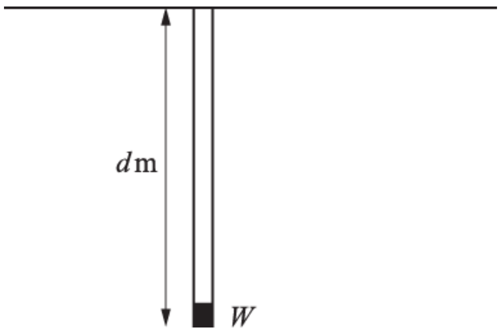
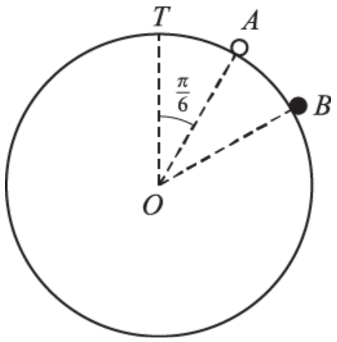


A small object W of weight 100 N is attached to one end of each of two parallel light elastic strings. One string is of natural length 0.4 m and has modulus of elasticity 20 N; the other string is of natural length 0.6 m and has modulus of elasticity 30 N. The upper ends of both strings are attached to a horizontal ceiling and W hangs in equilibrium at a distance d m below the ceiling (see diagram). Find d.

[5]

35



A fixed smooth sphere of radius 0.6 m has centre O and highest point T. A particle of mass m kg is released from rest at a point A on the sphere, such that angle TOA is $\frac{\pi}{6}$ radians. The particle leaves the surface of the sphere at B (see diagram).

(i) Show that $\cos TOB = \frac{\sqrt{3}}{3}$

[6]

(ii) Find the speed of the particle at *B*.

[2]

(iii) Find the transverse acceleration of the particle at *B*.

[2]

36 A car of mass 1400 kg is travelling on a straight horizontal road against a constant resistance to motion of 600 N. At a certain instant the car is accelerating at 0.3 m s^{-2} and the engine of the car is working at a rate of 23 kW.

(i) Find the speed of the car at this instant.

[3]

Subsequently the car moves up a hill inclined at 10° to the horizontal at a steady speed of 12 ms^{-1}. The resistance to motion is still a constant 600 N.

(ii) Calculate the power of the car's engine as it moves up the hill.

[3]

37 A and B are two points on a line of greatest slope of a plane inclined at 55° to the horizontal. A is below the level of B and AB = 4 m. A particle P of mass 2.5 kg is projected up the plane from A towards B and the speed of P at B is 6.7 m s^{-1}. The coefficient of friction between the plane and P is 0.15. Find

(i) the work done against the frictional force as P moves from A to B,

[3]

(ii) the initial speed of P at A.

[4]

38 The masses of two particles A and B are 4 kg and 3 kg respectively. The particles are moving towards each other along a straight line on a smooth horizontal surface. A has speed 8 m s^{-1} and B has speed 10 m s^{-1} before they collide. The kinetic energy lost due to the collision is 121.5 J.

(i) Find the speed and direction of motion of each particle after the collision.

[8]

(ii) Find the coefficient of restitution between A and B.

[2]

39 A cyclist travels along a straight horizontal road. The total mass of the cyclist and her bicycle is 80 kg and the resistance to motion is a constant 60 N.

(i) The cyclist travels at a constant speed working at a constant rate of 480 W. Find the speed at which she travels.

[3]

(ii) The cyclist now instantaneously increases her power to 600 W. After travelling at this power for 14.2 s her speed reaches 9.4 ms^{-1} . Find the distance travelled at this power.

[4]

40 A car of mass 1500 kg travels along a straight horizontal road with its engine working at a constant rate of P W. There is a constant resistance to motion of R N. Points A and B are on the road. At point A the car's speed is 16 ms^{-1} and its acceleration is 0.3875 ms^{-2}. At point B the car's speed is 25 ms^{-1} and its acceleration is 0.2 ms^{-2}. Find the values of P and R.

[6]

41 A small sphere of mass 0.2 kg is projected vertically downwards with a speed of 5 ms^{-1} from a height of 1.6 m above horizontal ground. It hits the ground and rebounds vertically upwards coming to instantaneous rest at a height of h m above the ground. The coefficient of restitution between the sphere and the ground is 0.7.

(i) Find h.

[4]

(ii) Find the magnitude and direction of the impulse exerted on the sphere by the ground.

[3]

(iii) Find the loss of energy of the sphere between the instant of projection and the instant it comes to instantaneous rest at height h m.

[3]

42

(i) A car of mass 800 kg is moving at a constant speed of 20 m s^{-1} on a straight road down a hill inclined at an angle α to the horizontal. The engine of the car works at a constant rate of 10 kW and there is a resistance to motion of 1300 N. Show that $\sin \alpha = \frac{5}{49}$.

[4]

(ii) The car now travels up the same hill and its engine now works at a constant rate of 20 kW. The resistance to motion remains 1300 N. The car starts from rest and its speed is 8 m s^{-1} after it has travelled a distance of 22.1 m. Calculate the time taken by the car to travel this distance.

[5]

43 *A* and *B* are two points on a line of greatest slope of a smooth inclined plane, with *B* a vertical distance of 8 m below the level of *A*. A particle of mass 0.75 kg is projected down the plane from *A* with a speed of 2 ms^{-1}. Find

(i) the loss in potential energy of the particle as it moves from *A* to *B*,

[2]

(ii) the speed of the particle when it reaches *B*.

[4]

44 The power developed by the engine of a car as it travels at a constant speed of 32 ms^{-1} on a horizontal road is 20 kW.

(i) Calculate the resistance to the motion of the car.

[3]

The car, of mass 1500 kg, now travels down a straight road inclined at 2° to the horizontal. The resistance to the motion of the car is unchanged.

(ii) Find the power produced by the engine of the car when the car has speed 32 ms^{-1} and is accelerating at 0.1 ms^{-2}.

[4]

45 A block is being pushed in a straight line along horizontal ground by a force of 18 N inclined at 15° below the horizontal. The block moves a distance of 6 m in 5 s with constant speed. Find

(i) the work done by the force,

[3]

(ii) the power with which the force is working.

[2]

46 A car of mass 1500 kg travels along a straight horizontal road. The resistance to the motion of the car is $kv^{\frac{1}{2}}$ N, where v ms^{-1} is the speed of the car and k is a constant. At the instant when the engine produces a power of 15 000 W, the car has speed 15 ms^{-1} and is accelerating at 0.4 ms^{-2}.

(i) Find the value of k.

[4]

It is given that the greatest steady speed of the car on this road is 30 ms^{-1}.

(ii) Find the greatest power that the engine can produce.

[3]

END OF QUESTION PAPER